MZUZU UNIVERSITY

FACULTY OF INFORMATION SCIENCE AND COMMUNICATIONS

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

BACHELOR OF SCIENCE IN INFORMATION AND COMMUNICATION TECHNOLOGY

END OF SEMESTER EXAMINATION

ICT 1103: COMPUTER AND COMMUNICATION TECHNOLOGY

TIME ALLOWED: 3 HOURS **DATE: FEBRUARY, 2018**

INSTRUCTIONS

i. This paper contains **Three** pages. Please check.
ii. There are **five questions** in this paper. Please check.
iii. Answer **ALL** questions.
iv. Marks for each question are indicated.

Question One

a) Describe **three** categories of input devices. **6 Marks**

b) Explain **two** reasons why the use of biometric devices are encouraged in security-sensitive systems. **4 Marks**

c) Explain **two** reasons why it is important to consider ergonomics before one buys a hardware. **4 Marks**

d) List **eight** input devices and **eight** output devices. **8 Marks**

e) Explain how IT has affected the following:

   a. Health. **2 Marks**

   b. Money. **2 Marks**

   c. Education. **2 Marks**

   d. Entertainment. **2 Marks**

f) Convert 7722_8 to Hex . **4 Marks**

g) Convert $6DF9_{16}$ to Binary. **3 Marks**

Question Two

a) With the aid of a diagram (s), describe

   i. AM. **4 Marks**

   ii. FM. **4 Marks**

b) Suppose you were hired as the Network Engineer for iTech company and you have been asked to design a Network for the company with the following requirements; there are 5 departments at the company (IT, Accounts, Administration, Human Resource, Production), Each department has 10 desktop computers, 10 Laptops and 10 Tablets, The company has 3 buildings in which the departments are housed.

   Design and organise a network using the provided requirements with focus on:

   i. Network topology and layout

   ii. Hardware with/and specifications

   iii. Software with/and specifications

   You may use a diagram(s) to explain. **12 marks**

c) Describe full-duplex and half-duplexcommunication. **4 marks**

d) What **five** factors would you consider before purchasing network cables? **5 Marks**

e) Mention **five** benefits of using database systems. **5 Marks**

Question Three

a) Facebook acquired WhatsApp at approximately 19 billion US dollars. With reference to e-commerce, can you justify this expenditure? **4 Marks**

b) Explain how the following are threats to computer systems:

   i. Computer crimes. **2 Marks**

   ii. Computer criminals. **2 Marks**

   iii. Errors and accidents. **2 Marks**

c) Discuss **five** components of security. **10 Marks**

d) Describe the SDLC **12 Marks**

Question Four

a) The manager of Takhuta Bakery believes that having a computer network and deploying information systems in his company can improve the overall performance of the bakery.

   i. Describe **three** types of information systems that can work better in this Bakery setup. **9 Marks**

   ii. Explain **four** cons that may arise in this company due to implementation of the information systems mentioned in 4a. **8 marks**

b) Explain **five** types of wireless communication media. **10 Marks**

Question Five

a) Describe **three** ways in which computers have developed. **6 marks**

b) Using a sketched diagram, describe a machine cycle. **8 Marks**

c) Explain **five** basic operations of a computer. **10 Marks**

d) Explain **five** ways to speed up data travelling between CPU and memory. **10 Marks**

e) Sketch the interior of a system unit and label :

   i. 3 slots. **3 Marks**

   ii. 3 chips. **3 Marks**

f) How does an Operating System achieve the following:

   i. Task management **2 Marks**

ii. Security management **2 Marks**

iii. CPU management **2 Marks**

End of question paper